

Lalithkumar Seenivasan

lalithjets@yahoo.com | [lalithjets.github.io](https://github.com/lalithjets)

 [lalithkumar-seenivasan](#) |  [lalithjets](#) |  [Google Scholar](#)

Residence: Baltimore, MD 21210, USA | Nationality: Singapore

EDUCATION AND TRAINING

- | | |
|--|---|
| • Postdoctoral Research Fellow
<i>Johns Hopkins University</i> | <i>Feb 2024 - Current</i>
Baltimore, USA |
| • Doctor of Philosophy (PhD), Biomedical Engineering
<i>National University of Singapore</i> | <i>2019 – 2024</i>
Singapore, Singapore |
| • B.Eng. (Hons) Electrical Engineering
<i>National University of Singapore</i> | <i>2016 – 2019</i>
Singapore, Singapore |
| • Diploma in Mechatronics
<i>Temasek Polytechnic</i> | <i>2010 – 2013</i>
Singapore, Singapore |

WORK EXPERIENCE

- | | |
|---|--|
| • Research Assistant
<i>The Chinese University of Hong Kong</i> | <i>Mar 2023 – Nov 2023</i>
Hong Kong, Hong Kong SAR |
| • Research Engineer
<i>National University of Singapore</i> | <i>Dec 2020 – Feb 2023</i>
Singapore, Singapore |
| • Engineering Manager
<i>Aitech Robotics and Automation Pte Ltd</i> | <i>Aug 2019 – Jul 2020</i>
Singapore, Singapore |
| • Software Engineer (Part-time), R&D Dept
<i>Aitech Robotics and Automation Pte Ltd</i> | <i>Jun 2016 – Jul 2019</i>
Singapore, Singapore |
| • Temp Technical Support Officer
<i>Temasek Polytechnic</i> | <i>Oct 2013 – Jan 2014</i>
Singapore, Singapore |

AWARDS AND HONORS

Academic Awards

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| • Outstanding Undergraduate Researcher Prize
<i>National University of Singapore</i> | <i>AY2017–2018</i> |
| • FoE 32nd Innovation & Research Award
<i>National University of Singapore</i> | <i>2018</i> |
| • Singapore Manufacturing Federation Metal, Machinery & Engineering Industry Group Project Prize
<i>Temasek Polytechnic, Singapore</i> | <i>2013</i> |
| • Commendation Award for Major Project
<i>Temasek Polytechnic, Singapore</i> | <i>2013</i> |
| • Director's List Award
<i>Temasek Polytechnic, Singapore</i> | <i>2013</i> |
| • Director's List Award
<i>Temasek Polytechnic, Singapore</i> | <i>2012</i> |

Conference and Journal Awards

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| • Distinguished Reviewer Bronze Level
<i>IEEE Transactions on Medical Imaging</i> | <i>2023 - 2024</i> |
| • STudent-Author Registration (STAR) Awards
<i>International Conference on Medical Image Computing and Computer Assisted Intervention</i> | <i>2023</i> |
| • Outstanding Reviewer Award
<i>International Conference on Medical Image Computing and Computer Assisted Intervention</i> | <i>2023</i> |

Leadership Awards

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|---|-------------|
| • Commander's Outstanding Performance Award
<i>National Service, Singapore Police Force</i> | <i>2015</i> |
| • Sword Of Merit, Officer Cadet Course
<i>National Service, Singapore Police Force</i> | <i>2014</i> |
| • CCA Merit Award in Leadership
<i>Temasek Polytechnic, Singapore</i> | <i>2013</i> |

Journal papers:

12. Soberanis-Mukul, R.D., Shankar, R., **Seenivasan, L.**, Porras, J.L., Ishii, M., & Unberath, M. (2025). The Interpretable Surgical Temporal Informer: Explainable Surgical Time Completion Prediction. *International Journal of Computer Assisted Radiology and Surgery*. [Special Issue: Information Processing in Computer-Assisted Interventions \(IPCAI\) 2025](#).
11. Shu, H., Liu, M., **Seenivasan, L.**, Gu, S., Ku, P.C., Knopf, J., Taylor, R., & Unberath, M. (2025). Seamless augmented reality integration in arthroscopy: a pipeline for articular reconstruction and guidance. *Healthcare Technology Letters*, 12(1), p.e12119. [Special Issue: MICCAI AE-CAI 2024](#).
10. Bai, L., Wang, G., Islam, M., **Seenivasan, L.**, Wang, A., & Ren, H. (2025). Surgical-VQLA++: Adversarial contrastive learning for calibrated robust visual question-localized answering in robotic surgery. *Information Fusion*, 113, p.102602.
9. Zhang, H., Killeen, B.D., Ku, Y.C., **Seenivasan, L.**, Zhao, Y., Liu, M., Yang, Y., Gu, S., Martin-Gomez, A., Osgood, G., & Unberath, M. (2024). StraightTrack: Towards mixed reality navigation system for percutaneous K-wire insertion. *Healthcare Technology Letters*, 11(6), pp.355-364. [Special Issue: MICCAI AE-CAI 2024](#).
8. Ding, H., **Seenivasan, L.**, Killeen, B.D., Cho, S.M., & Unberath, M. (2024). Digital twins as a unifying framework for surgical data science: the enabling role of geometric scene understanding. In *Artificial Intelligence Surgery*, 4(3), pp.109-138. [Invited submission to the Special Issue on Surgical Data Science - The Art of Data](#).
7. Islam, M., **Seenivasan, L.**, Sharan, S.P., Vieakash, V.K., Gupta, B., Glocker, B., & Ren, H. (2023). Paced-curriculum distillation with prediction and label uncertainty for image segmentation. *International Journal of Computer Assisted Radiology and Surgery*, 18(10), pp.1875-1883.
6. **Seenivasan, L.**, Islam, M., Xu, M., Lim, C.M., & Ren, H. (2023). Task-aware asynchronous multi-task model with class incremental contrastive learning for surgical scene understanding. *International Journal of Computer Assisted Radiology and Surgery*, 18(5), pp.921-928.
5. Pang, W., Islam, M., Mitheran, S., **Seenivasan, L.**, Xu, M., & Ren, H. (2022). Rethinking feature extraction: Gradient-based localized feature extraction for end-to-end surgical downstream tasks. In *IEEE Robotics and Automation Letters*, 7(4), 12623–12630. [Concurrently accepted and presented at IEEE International Conference on Robotics and Automation 2023](#).
4. **Seenivasan, L.**, Islam, M., Ng, C.F., Lim, C.M., & Ren, H. (2022). Biomimetic incremental domain generalization with a graph network for surgical scene understanding. *Biomimetics*, 7(2), p.68.
3. **Seenivasan, L.**, Mitheran, S., Islam, M., & Ren, H. (2022). Global-reasoned multi-task learning model for surgical scene understanding. In *IEEE Robotics and Automation Letters*, 7(2), 3858–3865. [Concurrently accepted and presented at IEEE International Conference on Robotics and Automation 2022](#).
2. **Seenivasan, L.**, Bai, F., Ji, M., Gu, X., Tse, Z.T.H., & Ren, H. (2020). Shape tracking of flexible morphing matters from depth images. *IEEE Sensors Journal*, 21(6), pp.8234-8244.
1. Xu, M., **Seenivasan, L.**, Yeo, L.L.L., & Ren, H. (2020). Stent Deployment Detection Using Radio Frequency-Based Sensor and Convolutional Neural Networks. *Advanced Intelligent Systems*, 2(10), p.2000092.

Conference papers:

6. Riviera, C., Byrd, G., Paul, W., Feldman, T., **Seenivasan, L.**, Booker, M., Holmes, E., Handelman, D., Kemp, B., Badger, A., Ding, H., Jatavallabhula, K.M., De Melo, C.M., Unberath, M., Chellappa, R. (2025). ConceptAgent: LLM-Driven Precondition Grounding and Tree Search for Robust Task Planning and Execution”, *International Conference for Robotics and Automation (ICRA)*.
5. **Seenivasan, L.**, Islam, M., Kannan, G., & Ren, H. (2023). Surgicalgpt: End-to-end language-vision gpt for visual question answering in surgery. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*. [Won the MICCAI Student-Author Registration \(STAR\) Award](#).
4. **Seenivasan, L.**, Islam, M., Krishna, A. K., & Ren, H. (2022). Surgical-vqa: Visual question answering in surgical scenes using transformer. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 33–43). Springer.
3. Lal, R., Swaminathan, R., **Seenivasan, L.**, Qiu, L., & Ren, H. (2021), August. ScoopNet: 6DOF pose estimation pipeline for origami-inspired worm robots. IN *2021 IEEE international conference on development and learning (ICDL)* (pp. 1-6). IEEE.
2. Islam, M., **Seenivasan, L.**, Ming, L. C., & Ren, H. (2020). Learning and reasoning with the graph structure representation in robotic surgery. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 627–636). Springer.
1. Ren, H., Chen, C.X., Cai, C., Ramachandra, K. & **Seenivasan, L.** (2017). Pilot study and design conceptualization for a slim single-port surgical manipulator with spring backbones and catheter-size channels. In *2017 IEEE International Conference on Information and Automation (ICIA)* (pp. 499-504). IEEE.

Book Chapters:

3. **Seenivasan, L.,** & Ren, H. (2025). Artificial intelligence to understand robotic surgery scenes. In *Handbook of Robotic Surgery* (pp. 179-188). Academic Press.
3. Cai, X., Cai, C.J., **Seenivasan, L.,** Tse, Z., & Ren, H. (2023). Untethered Soft Ferromagnetic Quad-Jaws Cootie Catcher with Selectively Coupled Degrees of Freedom. In *Deployable Multimodal Machine Intelligence: Applications in Biomedical Engineering* (pp. 347-376). Singapore: Springer Nature Singapore.
2. **Lalithkumar, S.,** Cai, X., Ramachandra, K., Wong, F., & Ren, H. (2020). Tendon routing and anchoring for cable-driven single-port surgical manipulators with spring backbones and luminal constraints. *Flexible Robotics in Medicine: A Design Journey of Motion Generation Mechanisms and Biorobotic System Development*, p.169.
1. Ramachandra, K., Cai, C.J., **Lalithkumar, S.,** Cai, X., Tse, Z.T., & Ren, H. (2020). Tunable stiffness using negative Poisson's ratio toward load-bearing continuum tubular mechanisms in medical robotics. In *Control Theory in Biomedical Engineering* (pp. 317-358). Academic Press.

Workshops and Symposium:

2. **Seenivasan, L.,** Kannan, G., Ren, H. (2023). Multimodal Reasoning and Language Prompting for Interactive Learning Assistant for Robotic Surgery. In: *International Joint Conference on Artificial Intelligence: Symposium on Multimodal Reasoning with LLM*.
1. Islam, M., **Seenivasan, L.,** Ren, H., & Glocker, B. (2021). Class-distribution-aware calibration for long-tailed visual recognition. In *UDL Workshop, International Conference on Machine Learning*.

Preprints:

9. **Seenivasan, L.,** Xu, J., Soberanis-Mukul, R.D., Ding, H., Byrd, G., Ku, Y.C., Porras, J.L., Ishii, M., & Unberath, M. (2025). Beyond Rigid AI: Towards Natural Human-Machine Symbiosis for Interoperative Surgical Assistance. Under review, *MICCAI*.
8. Sharma, G., **Seenivasan, L.,** Porras, J.L., Luu, C., Unberath, M., & Lee, E.J. (2025). LLapSA: Large Laparoscopic Surgical Assistant. Under review, *MICCAI*.
7. **Seenivasan, L.,** Cheng, J., Fang, J., Bai, J., Soberanis-Mukul, R.D., Mangulabnan, J.E., Vedula, S.S., Ishii, M., Hager, Gregory., Taylor, R.H., & Unberath, M. (2025). EndoTracker: Robustly Tracking Any Point in Endoscopic Surgical Scene.
6. Gomez, C., **Seenivasan, L.,** Zou, X., Yoon, J., Chu, S., Leong, A., Kramer, P., Ku, YC, Porras, J.L., Martin-Gomez, A., Ishii, M., & Unberath, M. (2025). Explainable AI for Automated User-specific Feedback in Surgical Skill Acquisition.
5. Soberanis-Mukul, R.D., Perez, A., **Seenivasan, L.,** Kwon, K., Zhang, H., Porras, J.L., Ishii, M., & Unberath, M. (2025). Scalable and Privacy-Preserving Digital Twin Foundation Model for the Operating Room Using Federated Learning.
4. Shen, Y., Ding, H., **Seenivasan, L.,** Shu, T., & Unberath, M. (2025). Position: Foundation Models Need Digital Twin Representations. *arXiv preprint arXiv:2505.03798*.
3. Perez, A., Zhang, H., Ku, Y.C., **Seenivasan, L.,** Soberanis, R.D., Porras, J.L., Day, R., Jopling, J., Najjar, P., & Unberath, M. (2025). Privacy-Preserving Operating Room Workflow Analysis using Digital Twins. *arXiv preprint arXiv:2504.12552*.
2. Shen, Y., Liu, B., Li, C., **Seenivasan, L.,** & Unberath, M. (2025). Online reasoning video segmentation with just-in-time digital twins. *arXiv preprint arXiv:2503.21056*.
1. Ding, H., **Seenivasan, L.,** Shu, H., Byrd, G., Zhang, H., Xiao, P., Barragan, J.A., Taylor, R.H., Kazanzides, P., & Unberath, M. (2024). Towards Robust Automation of Surgical Systems via Digital Twin-based Scene Representations from Foundation Models. In *arXiv preprint arXiv:2409.13107*.

SELECTED TALKS AND PRESENTATIONS

6. Multimodal Reasoning and Language Prompting for Interactive Learning Assistant for Robotic Surgery, International Joint Conference on Artificial Intelligence: Symposium on Multimodal Reasoning with LLM, Macau, 2023.
5. Gradient-Based Localized Feature Extraction for End-To-End Surgical Downstream Tasks, IEEE International Conference on Robotics and Automation, London, May. 2023.
4. Relational Reasoning in VQA Models for Visual Question Answering in Robotic Surgery, IEEE International Conference on Robotics and Automation, London, May. 2023.
3. Surgical-VQA: Visual Question Answering in Surgical Scenes Using Transformer, Medical Image Computing and Computer-Assisted Intervention (MICCAI), Singapore, Sep. 2022.

2. CLEARNESS: Cross-scaLe tEmporAl gRaph NETwork for Super-reSolutions, IUPESM World Congress on Medical Physics and Biomedical Engineering, Singapore, Jun. 2022.
1. Global-Reasoned Multi-Task Learning Model for Surgical Scene Understanding, IEEE International Conference on Robotics and Automation, Philadelphia, May. 2022.

LEADERSHIP EXPERIENCE

National Service, Singapore Police Force

- **Assistant Superintendent of Police (Reservist)** 2021 – 2023
Singapore Police Force
- **Inspector of Police (Reservist)** 2016 – 2021
Singapore Police Force
- **National Service Inspector** 2014 – 2016
Singapore Police Force

Co-Curricular Activities

- **Vice-Captain, Touch Rugby** 2018 – 2019
National University of Singapore Raffles Hall, Singapore
- **Captain, Touch Rugby** 2017 – 2018
National University of Singapore Raffles Hall, Singapore
- **Quartermaster, Temasek Polytechnic International Students Group** 2012 – 2013
Temasek Polytechnic, Singapore